

EPM213s - Electrical Measurements

Measurement of Power in Single Phase Circuits Using Dynamometers

Name: Abdelrahman Sherif Medhat

Id: 2300100

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Introduction

Measuring single-phase power is essential in both home and laboratory environments. Accurate power readings under different load situations are obtained using an electrodynamometer wattmeter.

Theory

True power is determined by the wattmeter through torque generation proportional to the instantaneous voltage and current product. The needle's displacement reflects the cycle's average power.

Apparatus and Circuit

- Electrodynamometer wattmeter

- Variable resistive load

- Voltmeter and Ammeter

- AC single-phase power source

Procedure

1. Insert the wattmeter into the circuit, placing the fixed coil in series and the moving coil parallel to the voltage source.

2. Adjust the load progressively and note down the voltage, current, and wattmeter needle movement.

3. Document all readings in a tabular format.

4. Determine the apparent power and compute the power factor.

Sample Data Table (Assumed Values)

Voltage (V)	Current (A)	Wattmeter Reading (W)	Apparent Power (VA)	Power Factor
230	2	400	460	0.87
230	3	690	690	1.00
230	4	780	920	0.85

Graph and Analysis

Graph: Power Factor vs Load Current

Analysis: An increase in current may lead to variations in the power factor based on the nature of the load (such as resistive or inductive).

Calculations and Observations

Real Power = Wattmeter Reading (W)

Apparent Power = Voltage × Current (VA)

Power Factor = Real Power / Apparent Power

Conclusion

This type of wattmeter ensures precise power measurements and aids in analyzing how the power factor shifts with various load types.



